



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., V, EEE, MECH A & B

Course Code & Title: OAN551 & Sensors and Transducers

Name of the Faculty member (s): Mr.S.Vijayakumar

Innovative Practice Description

- **Unit / Topic:** IV / Pressure- Diaphragm, Bellows, piezoelectric- Tactile sensors
- **Course Outcome:** CO : The students will be able to describe the basic principles of Optical, pressure, temperature sensors and various smart sensors
- **Topic Learning Outcome:** TLO: The students will be describe the application of different pressure and temperature sensors for process industry
- **Activity Chosen:** Jigsaw
- **Justification:**
- The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal; each person depends on all the others.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO2
CO3	2	2	1	1	2	2	2	1	1

- **PO / PSO mapped:**

Innovative practice	PO8	PO9
	2	2
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>
- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>



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Innovative Practice Description

- **Unit / Topic:** V/ Applications- Aerospace, Home appliances
- **Course Outcome:** CO : The students will be able to construct the DAQ systems with different sensors for real time applications
- **Topic Learning Outcome:** TLO: The students will be able to locate different type of sensors used in real life applications and paraphrase their importance
- **Activity Chosen:** Flipped class room
- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Students are asked to prepare more in depth than before. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PSO2
CO3	2	1	1	2	2	2	2	1

• **PO / PSO mapped:**

Innovative practice	PO8	PO9
	2	2
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students feel that they have improved self-learning. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. To make all students to participate I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.

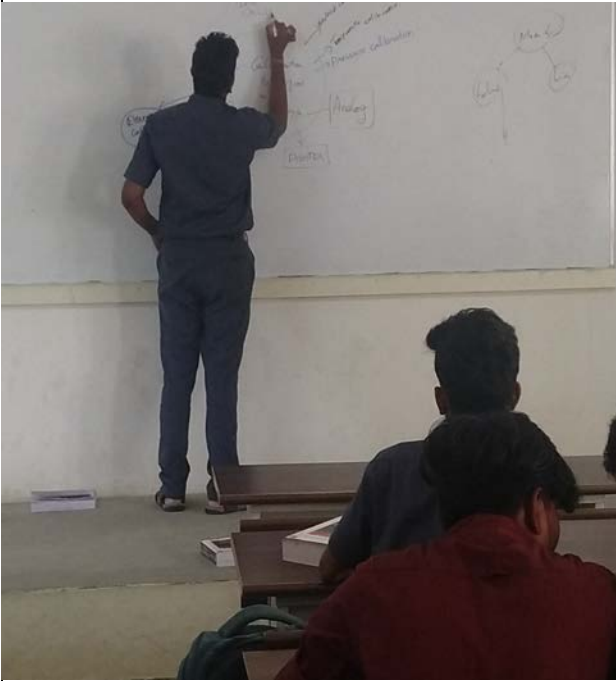
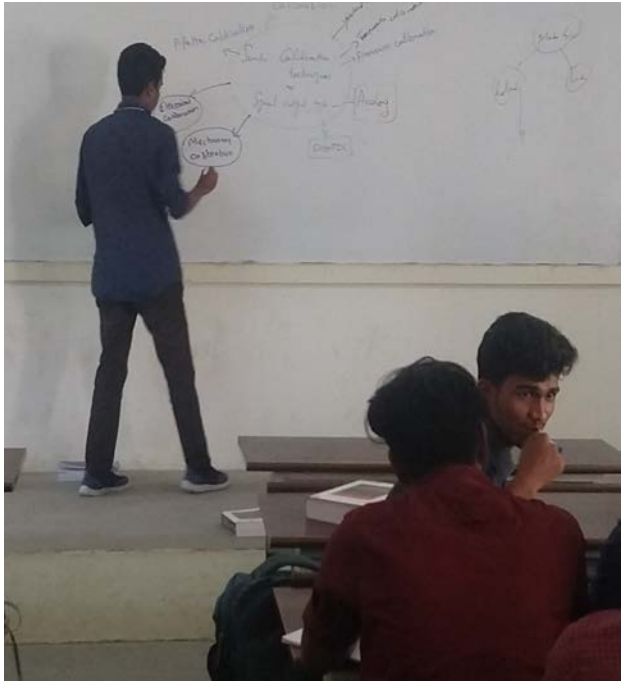
RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021 - 2022 (ODD Semester)
INNOVATIVE TEACHING METHOD PLAN

Degree, Semester & Branch: V Semester B.E.EEE, MECH A & B

Course Code & Title: OAN551 & Sensors and Transducers

Name of the Faculty member: Mr.S.Vijayakumar

UNIT I			
1	27.8.21	Static characteristics of Transducers	Exit slip
		953619114307 2-18/21 K. Rajasanthan <u>Static characteristics of Transducer:</u> It is used to measure the quantities such which are slowly varying with time or mostly constant. (i) Sensitivity (ii) Linearity (iii) Resolution (iv) Precision (Accuracy) (v) Span and Range (vi) Threshold (vii) Drift (viii) Stability (ix) Repeatability. Static error = true value of measured variable - Instrument reading.	<u>Static characteristic of transducer:-</u> (i) Sensitivity. (ii) Linearity. (iii) Resolution. (iv) precision (Accuracy) (v) Span and Range. (vi) Threshold. (viii) Drift. (ix) Stability (x) Repeatability. K. ELANGOVAN. 953619114016
2	01.9.21	Dynamic characteristics of Transducers	One minute paper
		<u>Dynamic characteristic</u> Elangoan.K 953619114016 (i) Dynamic Error (ii) Fidelity (iii) Speed of Response. (iv) Bandwidth.	<u>Dynamic characteristics</u> S. Jeyaswini 953619105010 Date: 28.08.21 1) Speed of response. 2) Zero order system 3) First order system 4) Measuring lag 5) Time delay 6) Fidelity 7) Dynamic error

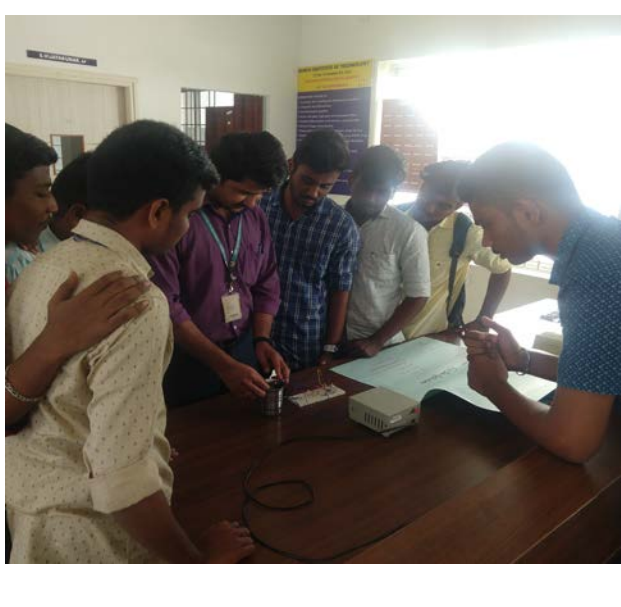
3	06.09.21	Sensor calibration techniques, Sensor output signal types & Revision	Mind maps
			

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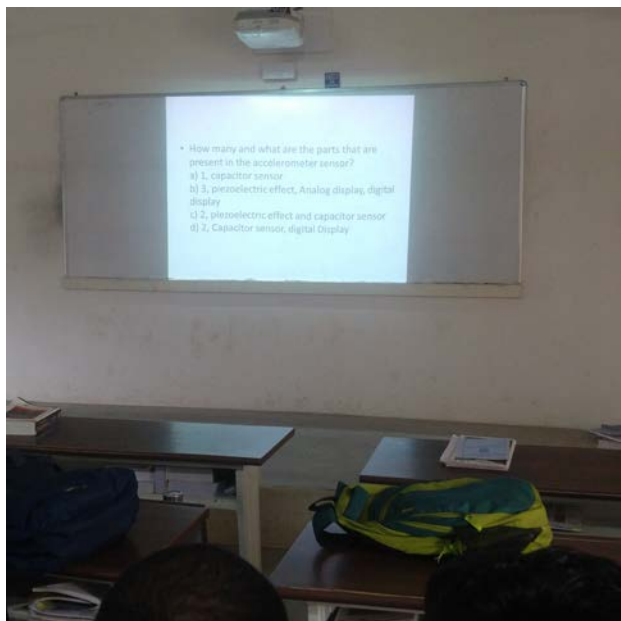
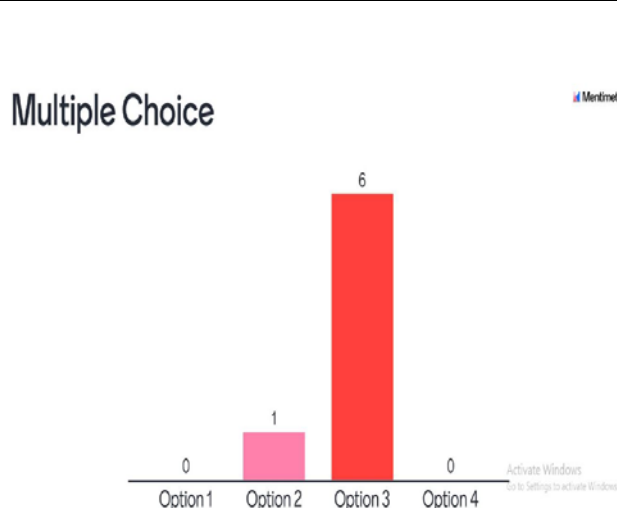
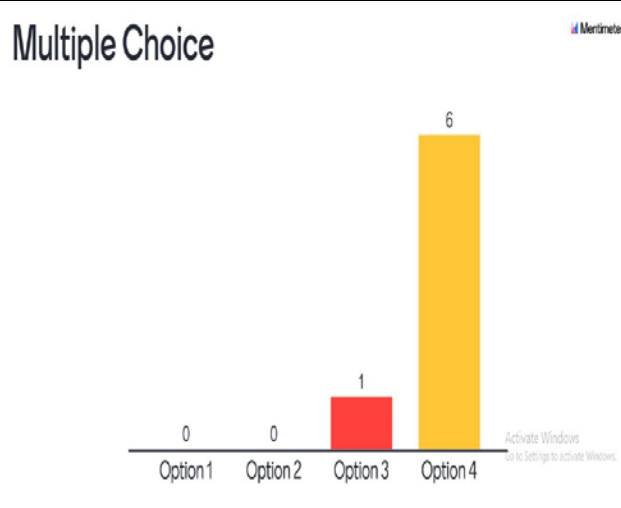
UNIT II			
1	08.9.21	Introduction to motion sensor-potentiometers	Live demonstration
			
2	29.9.21	GPS, Bluetooth	Brainstorming
			

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UNIT III			
1	08.10.21	Magnetic sensors- types, principle	Visible quiz
			
2	22.10.21	Heading Sensors- Compass, Gyroscope	Class Poll
			

UNIT III			
3	18.10.21	Hall effect	Note check
		<u>Hall effect transducers</u>, Vijay thirupathi In this transducers current and magnetic will pass as 90° due to this the hall voltage will produce. This voltage is known as Hall voltage H.V.	<u>Hall effect transducers</u>, 9536191110201 R. Raghav In this transducers current and magnetic will pass as 90° due to this the hall voltage will produce. This voltage is known as hall voltage.

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UNIT IV			
1	27.10.21	Photovoltaic, Photo resistive and LDR	Concept test
LDR light dependent Resistor works on the principle of photo conductivity. when the sunlight fall on the LDR the resistivity is decrease. when there is a dark mode the resistivity is increase. It is also called as photo-conductors. X K Kulkarni 95319105012 (EEE)		LDR Light depend resistance on that full measuring is used the street light and solar light are used.  ^{to} Photovoltaic The photovoltaic cell is on material is silicon that used solar panel to produce current also. 953619114029 R. Raghul	
2	12.11.21	Thermocouple, Acoustic Sensors	Open Book test
			

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UNIT V			
1	21.11.21	Filtering	Zero Minute speech
			
2	27.11.21	Home appliances	Case studies
			

